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PROGRAMMING WITH CPP 2

1. Scenario Analysis:

A bank account system is a good example of how Object-Oriented Programming (OOP) can be used in C++. The system can have a `BankAccount` class containing important information such as the account holder's name, account number, and balance. The class can also contain functions such as `deposit()`, `withdraw()`, and `displayBalance()`. When a customer deposits money, the deposit function increases the account balance. When money is withdrawn, the withdraw function checks whether sufficient funds are available before reducing the balance. Multiple objects can be created from the `BankAccount` class to represent different customers. OOP makes the program easier to organize, understand, and maintain because related data and functions are kept together. It also improves security by allowing sensitive information, such as the account balance, to be controlled through class methods. This scenario demonstrates how C++ OOP can be applied to create practical and reliable financial software.

2.concept Research:

Encapsulation is an important principle of Object-Oriented Programming (OOP) that combines data and the functions that operate on that data inside a class. It also protects data from direct access by using access modifiers such as `private`, `public`, and `protected`. For example, in a bank account system, the account balance can be kept `private` so that users cannot change it directly. Instead, public functions such as `deposit()` and `withdraw()` control how the balance is modified. Encapsulation improves security, prevents accidental changes, and makes programs easier to maintain. It helps programmers create organized, reliable, and reusable software.

3.Tool Practice:

I wrote and ran a C++ `Rectangle` class using Visual Studio Code. The class contained two private attributes, `length` and `width`, and a function to calculate the rectangle's area. I created an object from the class and used it to display the calculated area. This activity helped me understand how classes and objects work in Object-Oriented Programming. I also learned how encapsulation can be used to protect data inside a class. Running the program in Visual Studio Code helped me identify and correct errors in my code. Overall, the exercise improved my confidence in writing, testing, and understanding basic C++ OOP programs.

4. OOP Diagram:

The C++ class structure diagram shows how a class organizes data and functions in Object-Oriented Programming. The class contains attributes, which store information, and methods, which perform actions. For example, a Student class may contain attributes such as name, student number, and marks. Its methods could include `displayDetails()` and `calculateGrade()`. Objects can then be created from the Student class to represent individual students. This structure makes programs easier to organize, reuse, and maintain. Encapsulation can also protect important data by controlling access through private and public members. Overall, the diagram demonstrates how classes and objects work together in C++ programming.